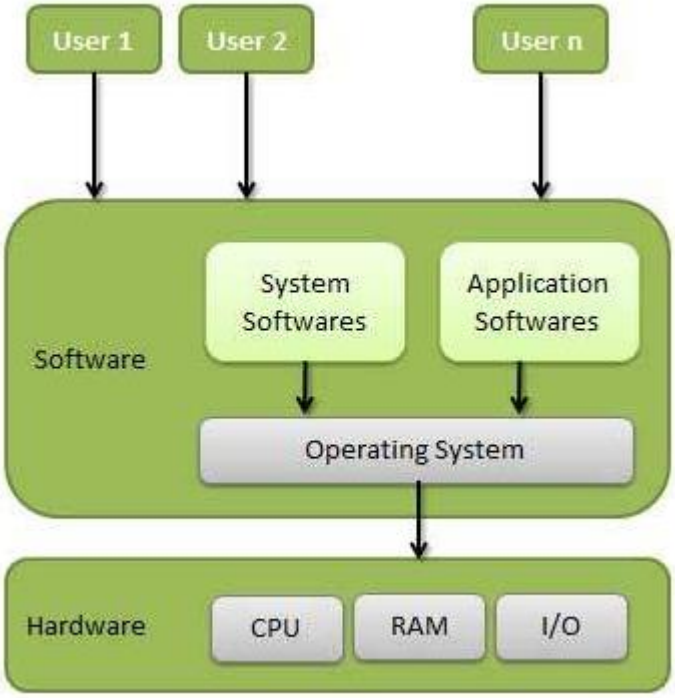
OPERATING SYSTEMS

OVERVIEW

An operating system (OS) is a collection of software that manages computer hardware and controls the allocation of resources and services such as memory, processors, devices and information. It is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.



TASKS/FUNCTIONS

There are 4 broad tasks performed by an operating system.

Process Management

A process is an executing program. It has its code, data, a certain set of resources allowed to it, and one or more flows of execution through the code. The OS manages the allocation of resources to these processes.

In multiprogramming environment, the OS decides which process gets the processor when and how much time it has that access. This function is called process scheduling.

Operating System does the following activities for processor management:

- Keeps track of the processor and status of processes. The program responsible for this task is known as traffic controller.

- It allocates the processor (CPU) to a process.
- It de-allocates the processor when it is no longer required.

Memory Management

The operating system handles or manages primary memory. It keeps track of each and every memory location, whether it is allocated to some process or it is free. Whenever some memory gets freed or unallocated the OS correspondingly updates the status.

It checks how much memory is to be allocated to processes.

It decides which process will get memory at what time.

File System Management

Computers process information that must be transmitted, processed, or stored. File systems are organized into directories for easy navigation and usage. The operating system does the following activities for file management:

- Keeps track of information, its location, uses, status etc. The collective facilities are often known as file system.
- Decides who gets which resources, allocates them and de-allocates them as need arises.

Device Management

Information is sent through a computer's input and output devices. The OS tries to manage these devices in a manner that makes them efficiently shared among all processes requiring them. It manages device communication via their respective drivers. The operating system does the following activities for device management:

- Keeps track of all devices. The program responsible for this task is known as the I/O controller.
- It decides which process gets the device when and for how long.
- Allocates the device in the efficient way.
- De-allocates devices when no longer required.

Other Important Activities

The following are some of the other important activities that the Operating System does:

Security: Considering a computer system having multiple users and concurrent execution of multiple processes, the various processes must be protected from each other's activities.

Protection refers to a mechanism or a way to control the access of programs, processes, or users to the resources defined by computer systems. By use of passwords, the OS can prevent unauthorized access to programs and data.

Communication: In case of distributed systems which are a collection of processors that do not share memory, peripheral devices, or a clock, the operating system manages communications between processes. Multiple processes link with one another through communication lines in the network. The OS handles routing and connection strategies.

Control over system performance: recording delays between request for a service and response from the system

Job accounting: keeping track of time and resources used by various jobs and users.

Error detecting aids: production of dumps, traces, error messages and other debugging and error detecting aids.

Coordination between other software and users: coordination and assignment of compilers, interpreters, assemblers and other software to the various users of the computer systems.

TYPES OF OPERATING SYSTEMS

Within the broad family of operating systems, there are generally four types, categorized based on the types of computers they control and the sort of applications they support. The categories are:

Batch Operating Systems

The users of batch operating systems do not interact with the computer directly. Each user prepares his job on an off-line device like punch cards and submits it to the computer operator. To speed up processing, jobs with similar needs are batched together and run as a group.

Time-sharing Operating Systems

Time sharing is a technique which enables many people, located at various terminals, to use a particular computer system at the same time. Time-sharing or multitasking is a logical extension of multiprogramming. The processor's time is shared among multiple users simultaneously thus the term time-sharing.

Multiple jobs are executed by the CPU by switching between them, but the switches occur so fast and frequently that the user can receive an immediate response.

Distributed Operating System

Distributed systems use multiple central processors to serve multiple real time application and multiple users. Data processing jobs are distributed among the processors according to which one can perform each job most efficiently. This OS distributes computation logics among several physical processors. The processors do not share memory or a clock. Instead, each processor has its own local memory.

The OS manages the communications between the processors, which communicate with each other through various communication lines.

The advantages of distributed systems include:

- With resource sharing facility a user at one site may be able to use the resources available at another.
- If one site fails in a distributed system, the remaining sites can potentially continue operating.
- Reduction of the load on the host computer.

- Reduction of delays in data processing.

Network Operating System

A network Operating System runs on a server and provides the server the capability to manage data, users, groups, security, applications, and other networking functions. The primary purpose of the network operating system is to allow shared file and printer access among multiple computers in a network, typically a local area network (LAN). Examples of network operating systems are Microsoft Windows Server 2003, Microsoft Windows Server 2008, Novell NetWare etc.

The advantages of network operating systems include:

- Centralized servers are highly stable.
- Security is server managed.
- Upgrades to new technologies and hardware can be easily integrated into the system.
- Remote access to servers is possible from different locations and types of systems.

The disadvantages of network operating systems are:

- High cost of buying and running a server.
- Dependency on a central location for most operations.
- Regular maintenance and updates are required.

Real Time Operating System

Real time system is defined as a data processing system in which the time interval required to process and respond to inputs is so small that it controls the environment. Real time processing is always on line whereas on line system need not be real time. The time taken by the system to respond to an input and display of required updated information is termed as response time. Real-time systems are used when there are rigid time requirements on the operation of a processor or the flow of data and real-time systems can be used as a control device in a dedicated application. The Operating system must guarantee response to events within fixed periods of time to ensure correct performance. For example Scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, and home-appliance controllers, Air traffic control system etc. In such systems, Operating Systems typically read from and react to sensor data.

There are two types of real-time operating systems:

- Hard real-time systems: hard real-time systems guarantee that critical tasks complete on time. In hard real-time systems secondary storage is limited or missing with data stored in ROM. In these systems virtual memory is almost never found.
- Soft real-time systems: Soft real time systems are less restrictive. Critical real-time task gets priority over other tasks and retains the priority until it completes. Soft real-time systems have limited utility than hard real-time systems. For example, Multimedia, virtual reality, Advanced Scientific Projects like undersea exploration and planetary rovers etc.

Single-user, single task

As the name implies, this operating system is designed to manage the computer so that one user can effectively do one thing at a time. The Palm OS for Palm handheld computers is a good example of a modern single-user, single-task operating system.

Single-user, multi-tasking

This is the type of operating system most people use on their desktop and laptop computers today. Microsoft's Windows and Apple's MacOS platforms are both examples of operating systems that will let a single user have several programs in operation at the same time. For example, it's entirely possible for a Windows user to be writing a note in a word processor while downloading a file from the Internet while printing the text of an e-mail message.

Multi-user

A multi-user operating system allows many different users to take advantage of the computer's resources simultaneously. The operating system must make sure that the requirements of the various users are balanced, and that each of the programs they are using has sufficient and separate resources so that a problem with one user doesn't affect the entire community of users. Unix, VMS and mainframe operating systems, such as MVS, are examples of multi-user operating systems.

It's important to differentiate between multi-user operating systems and single-user operating systems that support networking. Windows 2000 and Novell Netware can each support hundreds or thousands of networked users, but the operating systems themselves aren't true multi-user operating systems. The system administrator is the only "user" for Windows 2000 or

Netware. The network support and all of the remote user logins the network enables are, in the overall plan of the operating system, a program being run by the administrative user.

EXAMPLES OF OPERATING SYSTEMS

Mac OS by Apple

Linux by Community

Android by Google

UNIX

Features

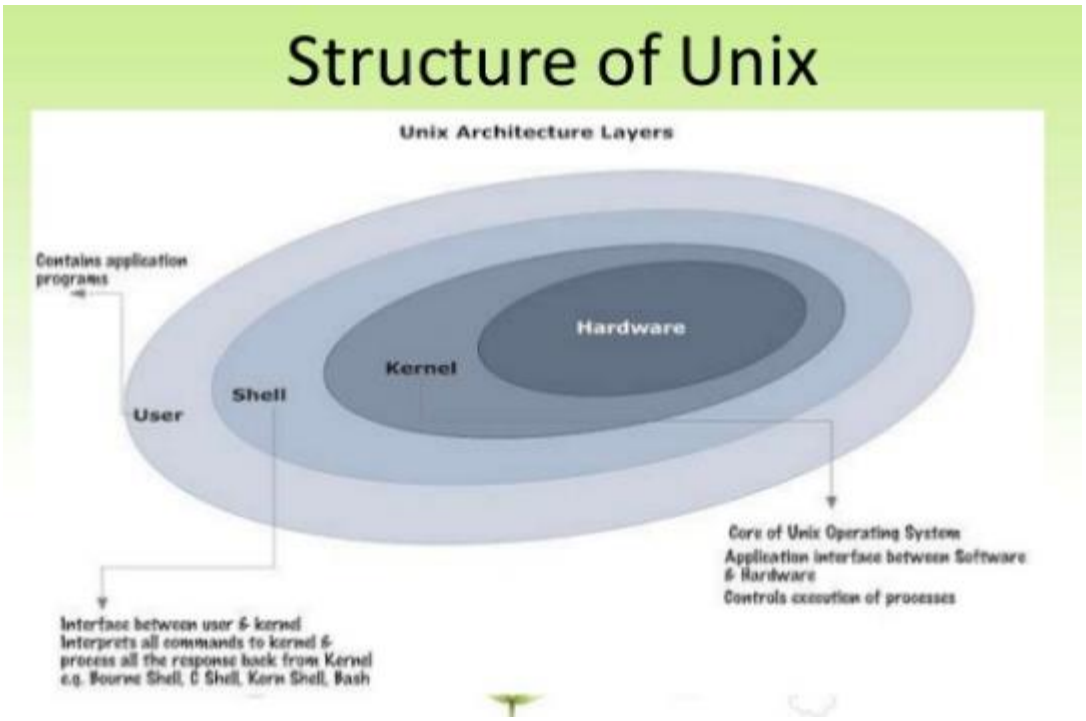
Multi user, Multitasking: Multiple users access the system by connecting to points known as terminals. Several users can run multiple programs or processes simultaneously on one system.

Portability: UNIX uses a high-level language that is easy to comprehend, modify and transfer to other machines, which means you can change language codes according to the requirements of new hardware on your computer. You, therefore, have the flexibility to choose any hardware, modify the UNIX codes accordingly and use UNIX across multiple architectures.

The Kernel: The kernel manages the applications and peripherals on a system. A user communicates with the UNIX shell which translates to the kernel.

The UNIX shell: is a program that gives and displays your prompts and, in conjunction with the kernel, executes your commands. The shell even maintains a history of the commands you enter, allowing you to reuse a command by scrolling through your history of commands.

Files and Processes: all the functions in UNIX involve either a file or a process. Processes are executions of programs, while files are collections of data created by you. Files may include a document, programming instructions for the system or a directory. UNIX uses a hierarchical file structure in its design that starts with a root directory--signified by the forward slash (/). The root is followed by its subdirectories, as in an inverted tree, and ends with the file. In the example "/Demand/Articles/UNIX.doc," the main directory "Demand" has a subdirectory "Articles," which has a file "UNIX.doc."



File Management in UNIX

Ordinary Files: files on the system that contain data, text or program instructions.

Directories: store both special and ordinary files. These are equivalent to folders in Windows.

Special Files: provide access to hardware such as hard drives, CD-ROM drives, modems and Ethernet adapters. Other special files are similar to aliases or shortcuts and enable you to access a single file using different names.

MAC OS

A series of GUI based OS developed by Apple to run on its computers.

Advantages of Mac OS

Easy to use: much easier to use than any other operating system.

Integration with Apple and other products: will automatically recognize and work with any other Apple product, such as iPhone, Cinema Display, Airport and iPad. Additional drivers are rarely needed for non-Apple products.

Mac OS can run Windows at the same time. Since it runs on Intel Chips, windows can run concurrently, with Boot Camp or Parallel software.

Reliability: very stable as it was designed from scratch and Apple controls production from start to finish. Every part is designed and tested to work together.

Design: engineered to take full advantage of what the hardware is capable of.

Disadvantages of Mac OS

Gaming: Boot camp and Parallels will allow you to run games natively in Windows. There is a very limited number of games produced for Mac OS.

Price: Macs cost more than machines that run on other operating systems.

Fewer Software Options: not all applications are yet available for Mac unlike Windows.

Cannot be upgraded /customized: there are upgrade options when you buy a Mac but you cannot mix and match components. The easiest way to upgrade is to buy a new one.

LINUX OS

Characteristics

Portable: can work on different types of hardware in the same way. Linux kernel and application programs support their installation on any kind of hardware platform.

Open source: the source code is freely available and is a community based development project. Multiple teams work in collaboration to enhance its capability.

Multi-User: multiple users can access system resources like memory/ ram / application programs at the same time.

Multiprogramming: multiple applications can run at the same time.

Hierarchical file system: It provides a standard file structure in which system files and user files are arranged.

Shell: it provides a special interpreter program which can be used to execute commands

Security: it provides user security using passwords, controlled access to specific files, etc.

Advantages of Linux OS

- It's a free operating system software unlike windows which has to be purchased.
- Since it is open source, anyone with programming knowledge can modify it.
- Once installed in the system, no antivirus is required as it is very secure.
- With its upgrades, Linux becomes even more secure and robust.
- Linux based servers could run non-stop without a reboot for years.

Special features

- **Live CD/USB:** Almost all Linux distributions have Live CD/USB feature by which user can run/try the OS even without installing it on the system.

- **Graphical user interface (X Window System):** People think that Linux is a command line OS, somewhere its true also but not necessarily, Linux have packages which can be installed to make the whole OS graphics based as Windows.
- **Support's** most national **or customized keyboards:** Linux is used worldwide and hence available in multiple languages, and supports most of their custom national keyboards.
- **Application Support:** Linux has its own software repository from where users can download and install thousands of applications just by issuing a command in Linux Terminal or Shell. Linux can also run Windows applications if needed.

MICROSOFT WINDOWS

Windows 10, Windows 8, Windows 7, Windows Vista, Windows XP, Windows 2000, Windows NT, Windows ME, Windows 98, Windows 95, MS-DOS

MS-DOS

Single user, single task

Machine dependent

Program control

Peripheral management